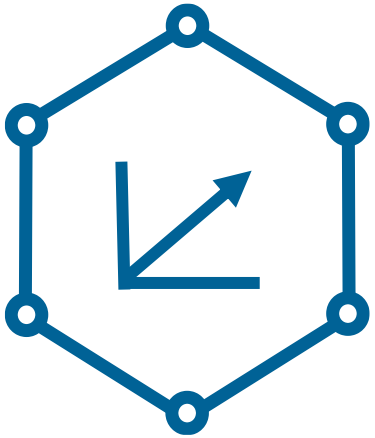


Knowledge Graph Embeddings

TransE: Learning

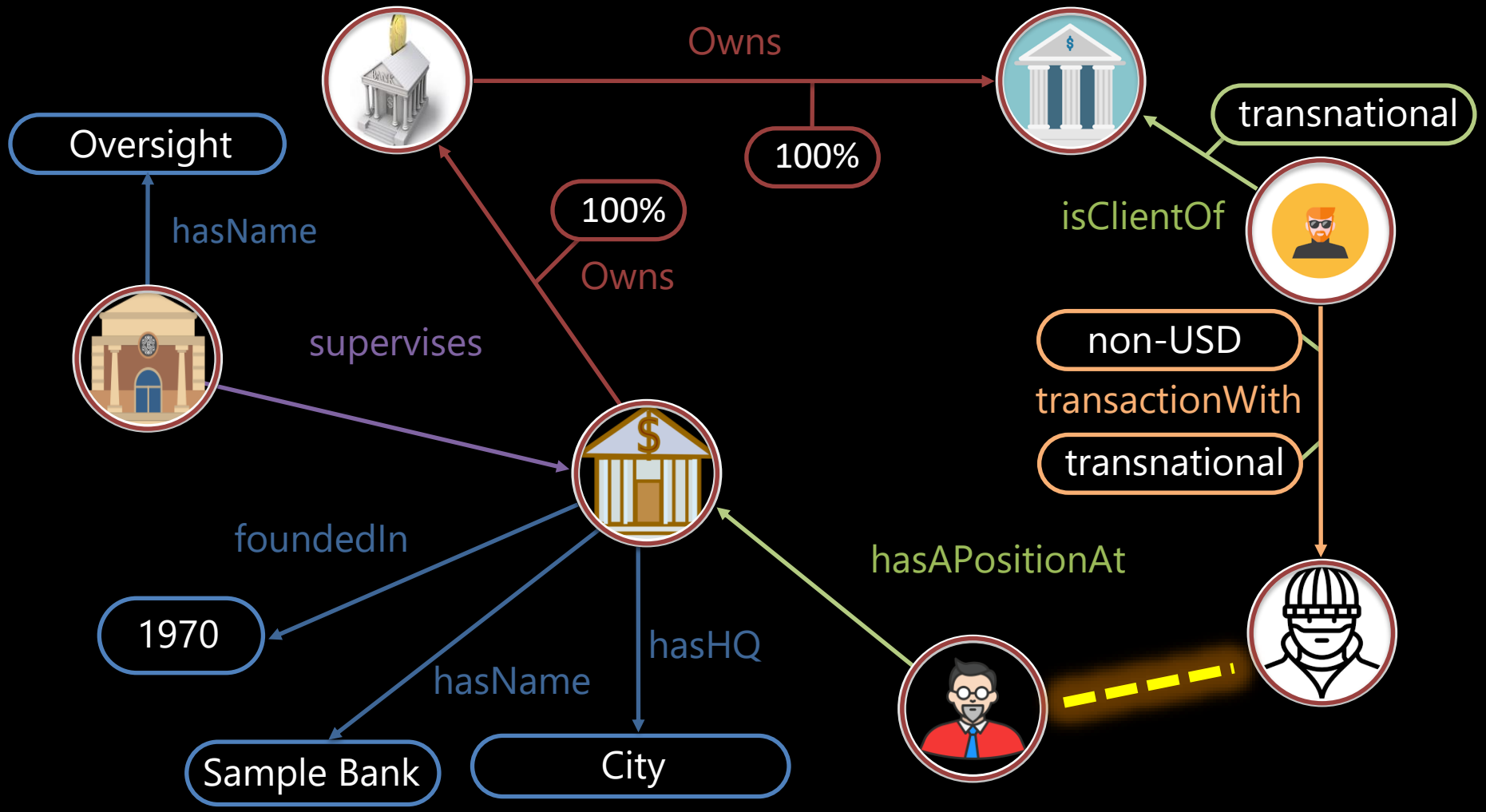
Emanuel Sallinger

Main Parts

1. **Score** Function
   how likely does a particular edge hold?

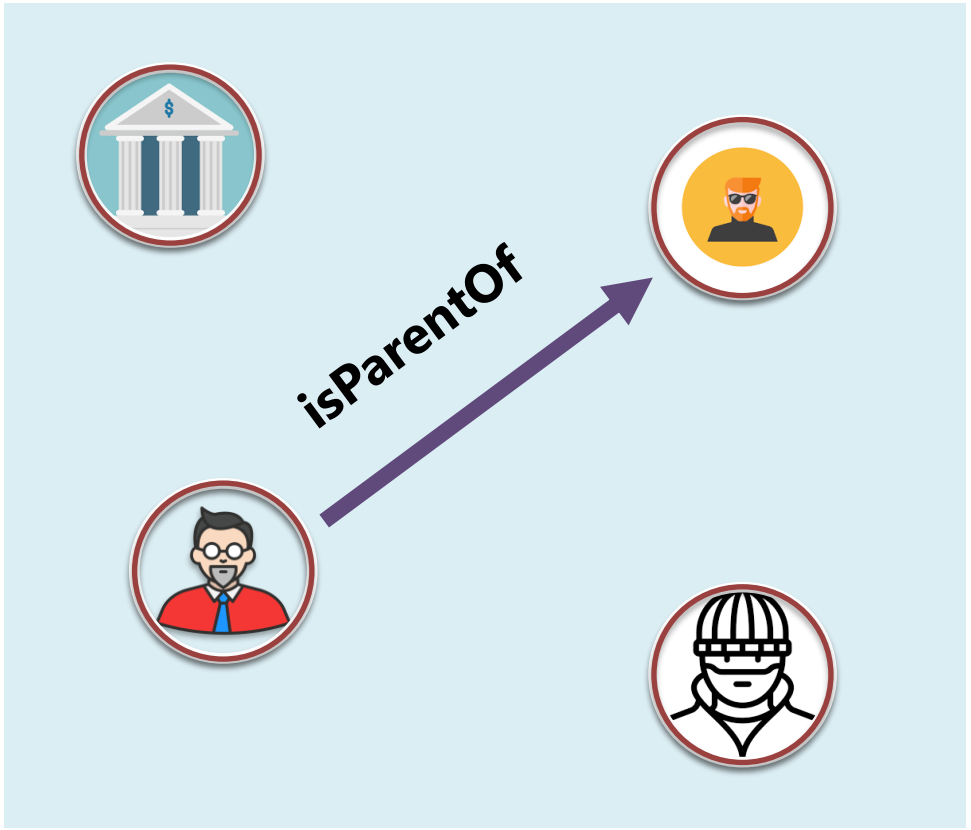
2. **Loss** Function
   what is our objective for optimization?

3. **Learning** Algorithm
   integrating the loss and score function


Owns
100%
Oversight
hasName
100%
Owns
transnational
isClientOf
supervises
non-USD
transactionWith
transnational
foundedIn
1970
hasName
hasHQ
hasAPositionAt
Sample Bank
City

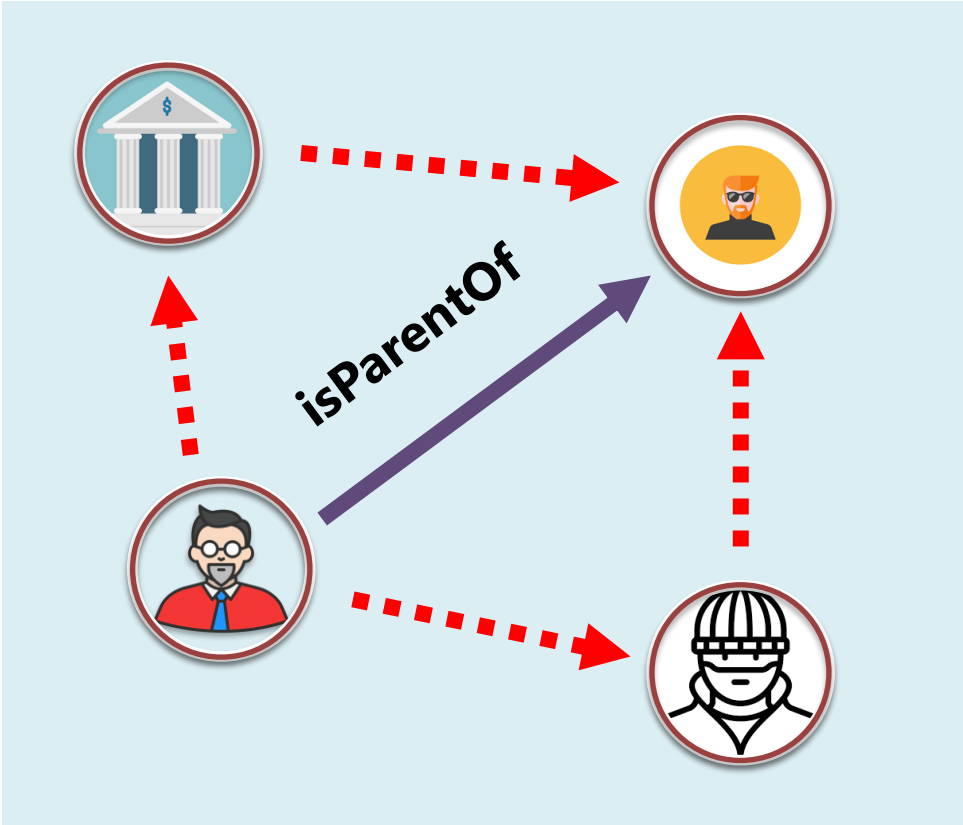
isParentOf holds in our KG

E



isParentOf holds in our KG

E

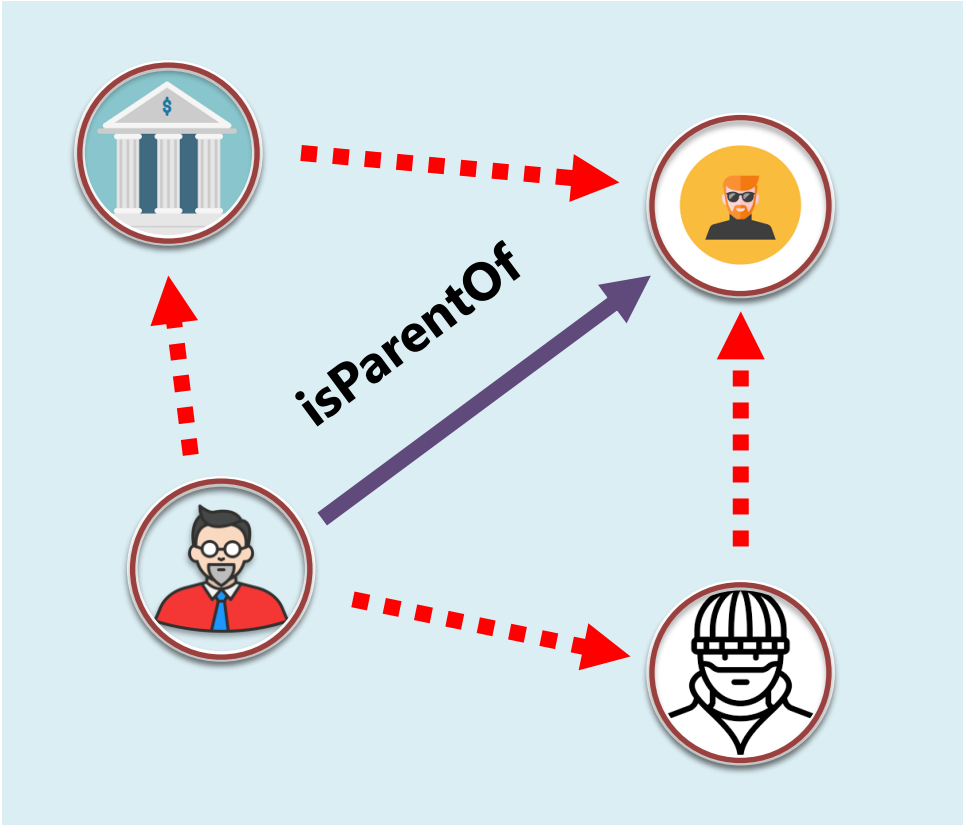


assume **isParentOf** is false

E'

isParentOf holds in our KG

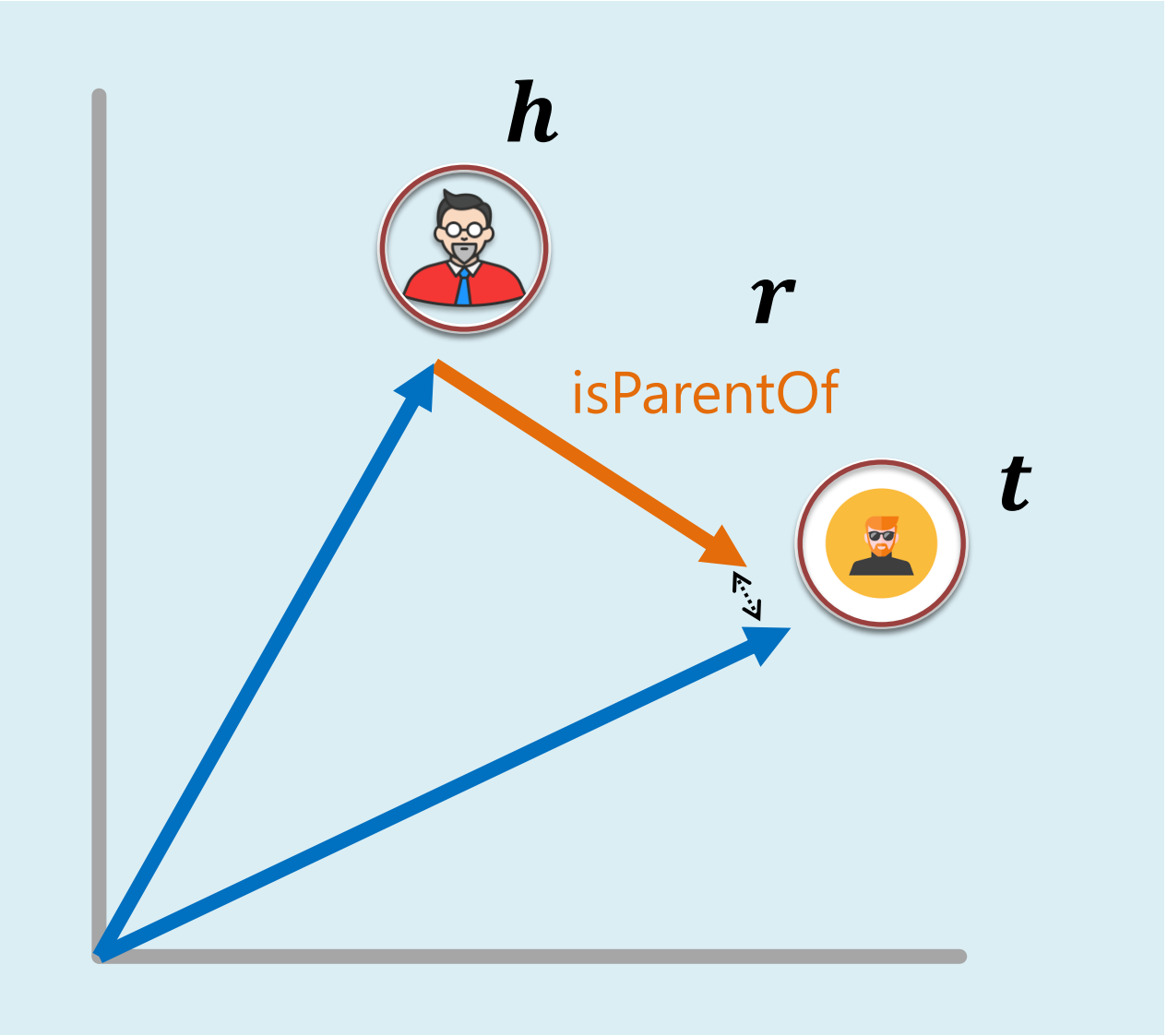
E

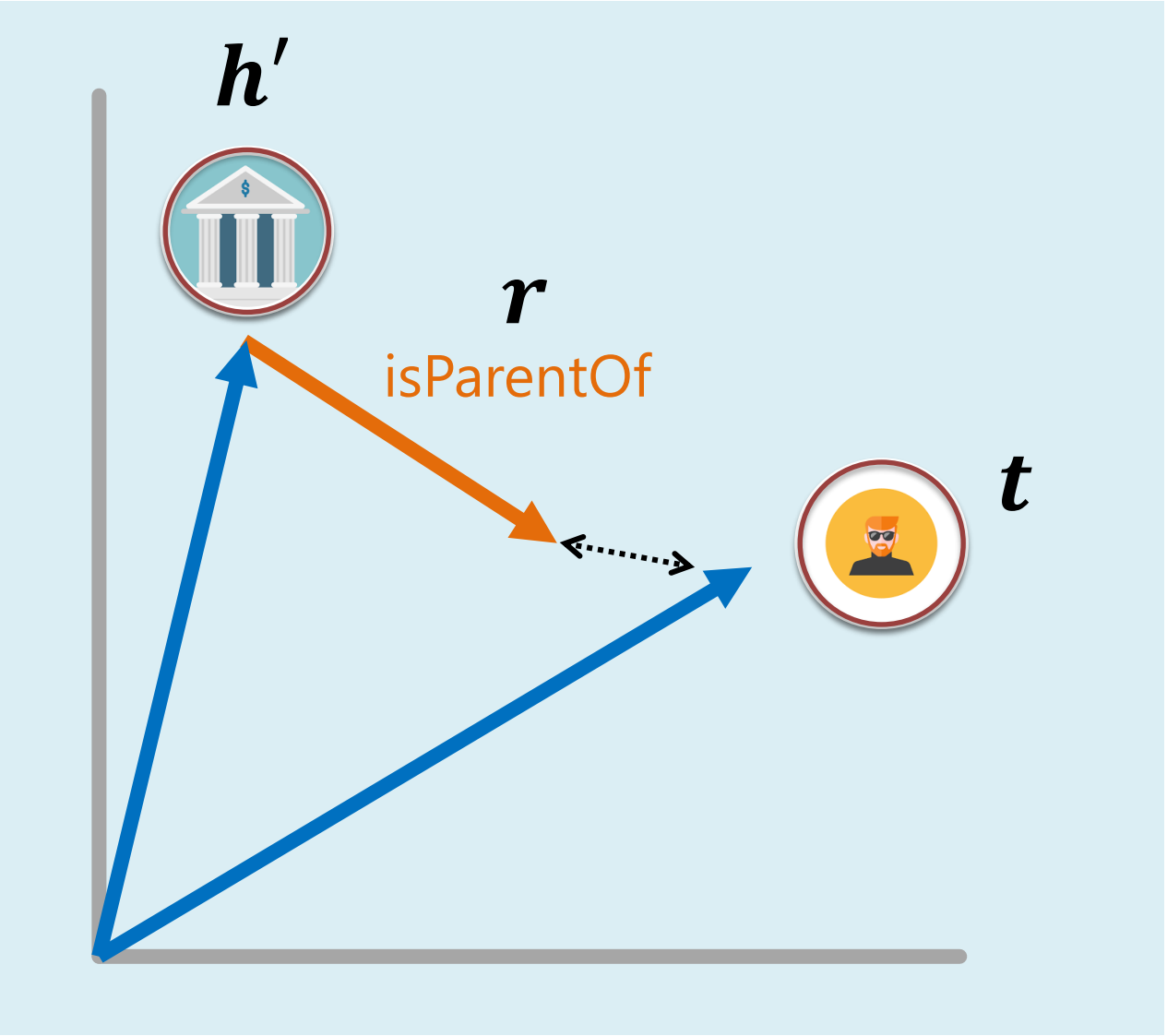


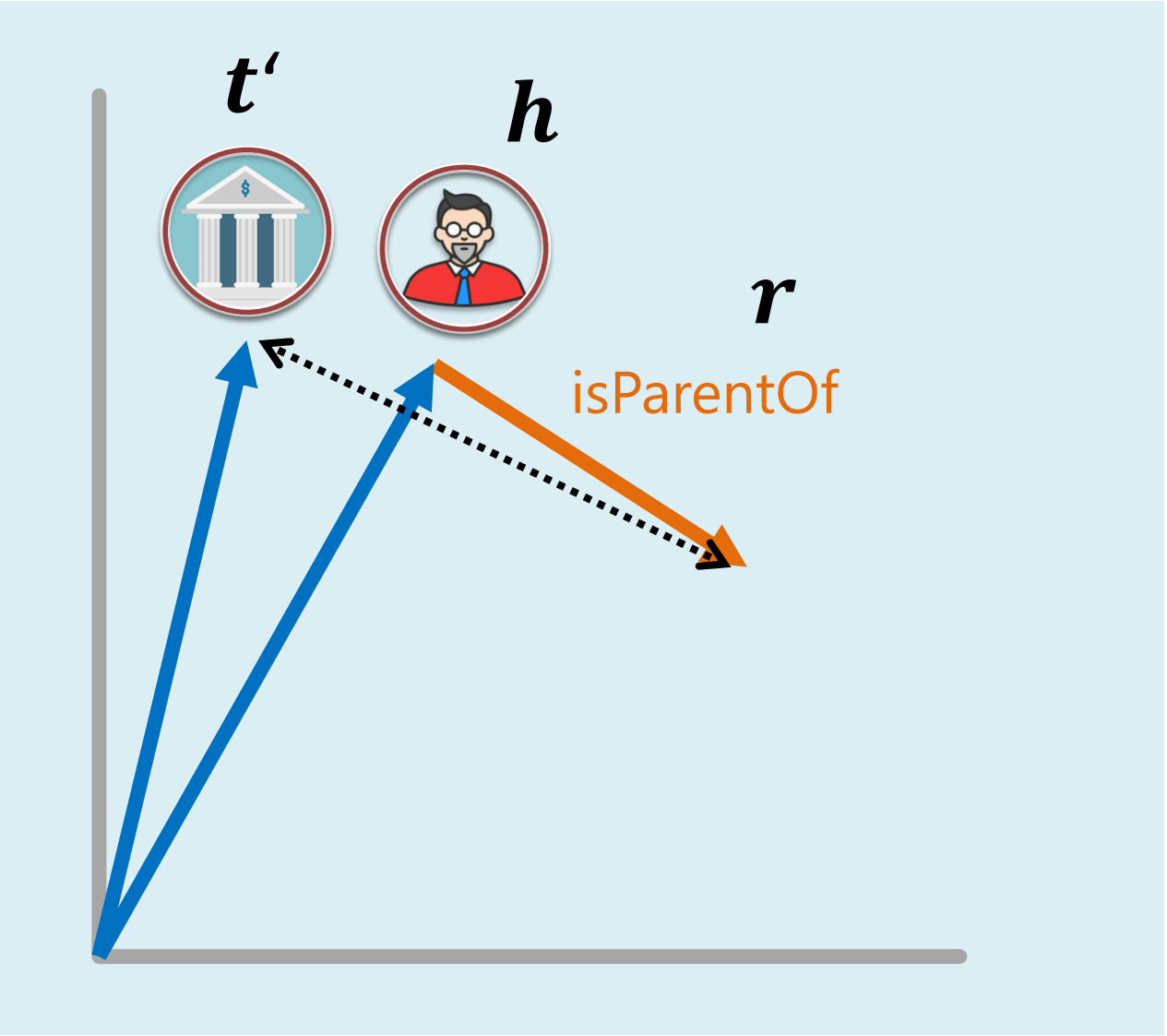
assume **isParentOf** is false

E'

$$E'_{(h,r,t)} = \{(h', r, t) \mid h' \in V\} \cup \{(h, r, t') \mid t' \in V\}$$

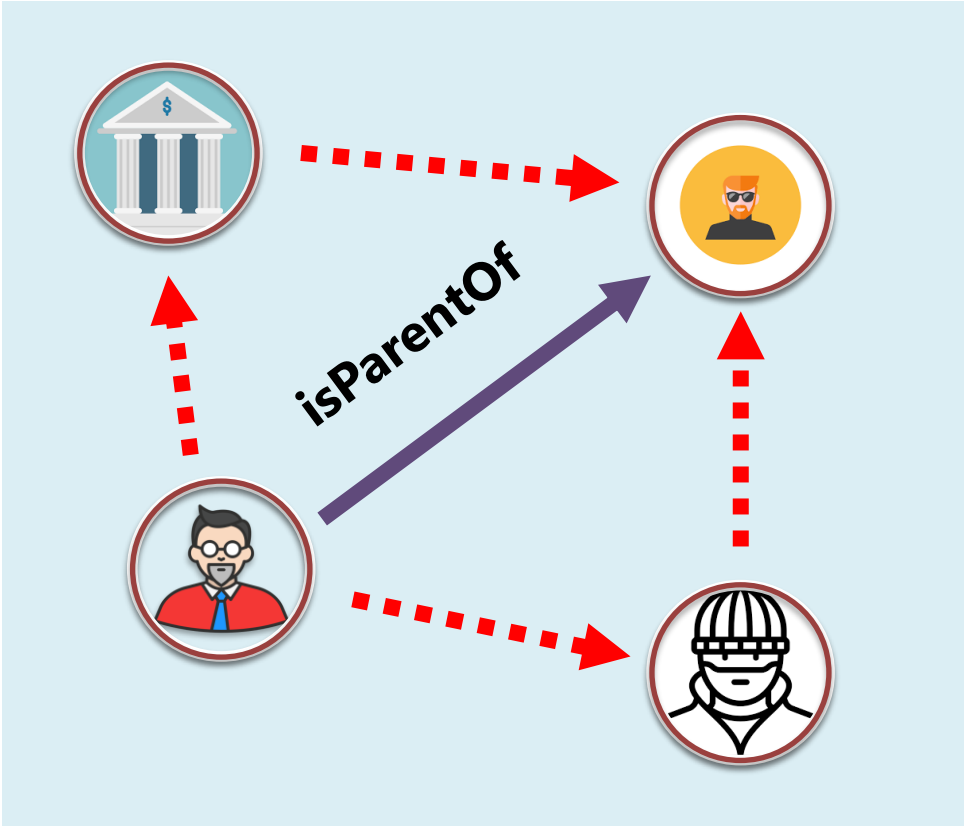

h
r
isParentOf
t


h'
r
isParentOf
t


t'
h
r
isParentOf

isParentOf holds in our KG

E



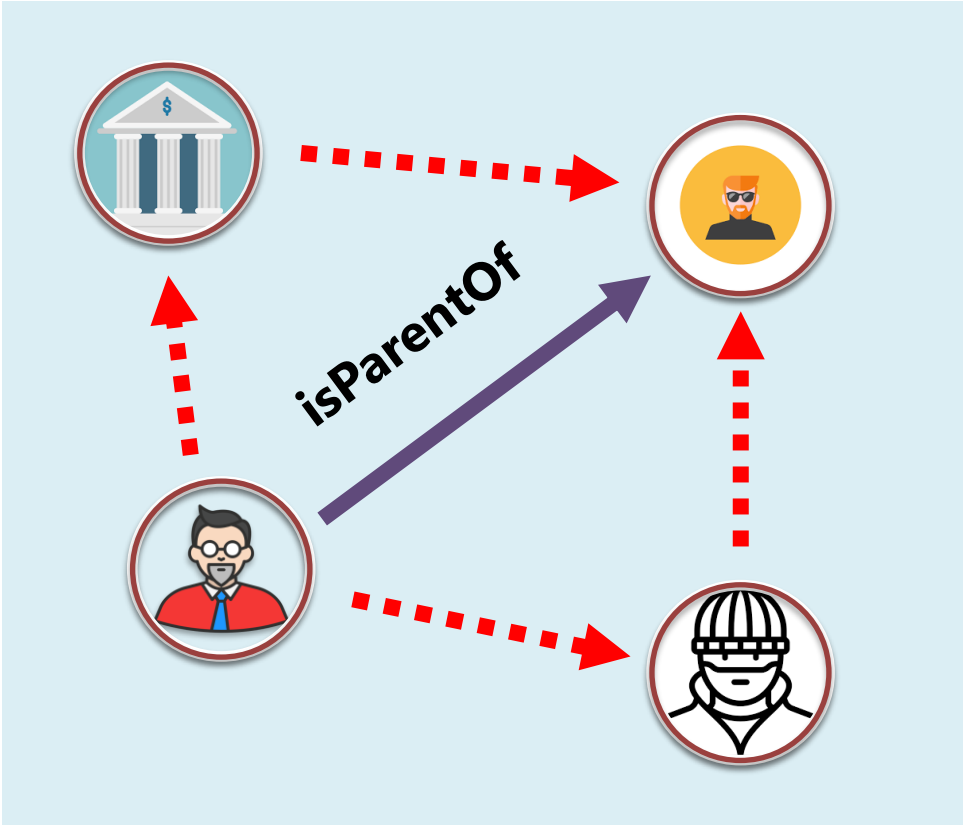
assume **isParentOf** is false

E'

$$\mathcal{L} = \sum_{(h,r,t)\in E} \sum_{(h',r,t')\in E'_{(h,r,t)}} [\gamma + f(h,r,t) - f(h',r,t)]_+$$

isParentOf holds in our KG

E

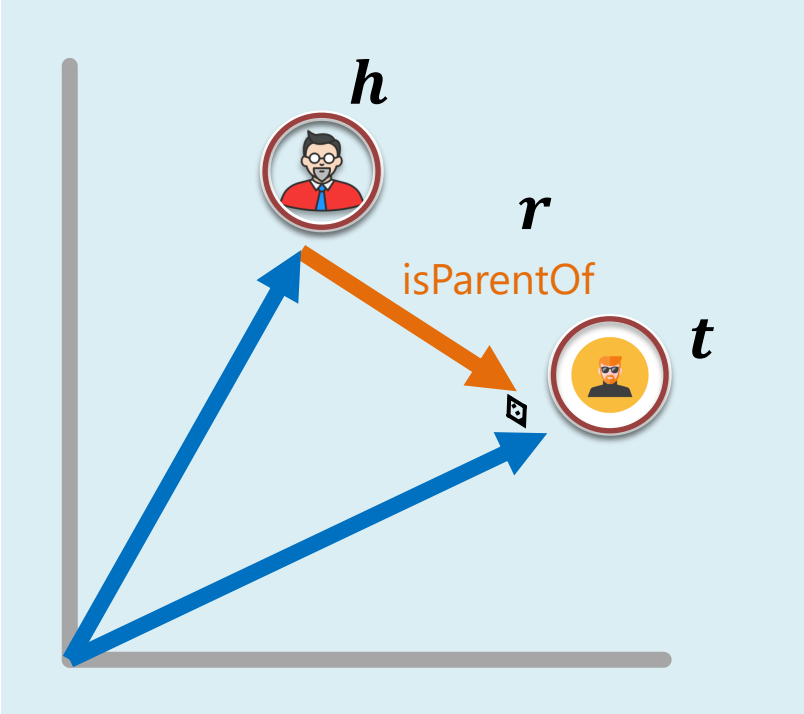


assume **isParentOf** is false

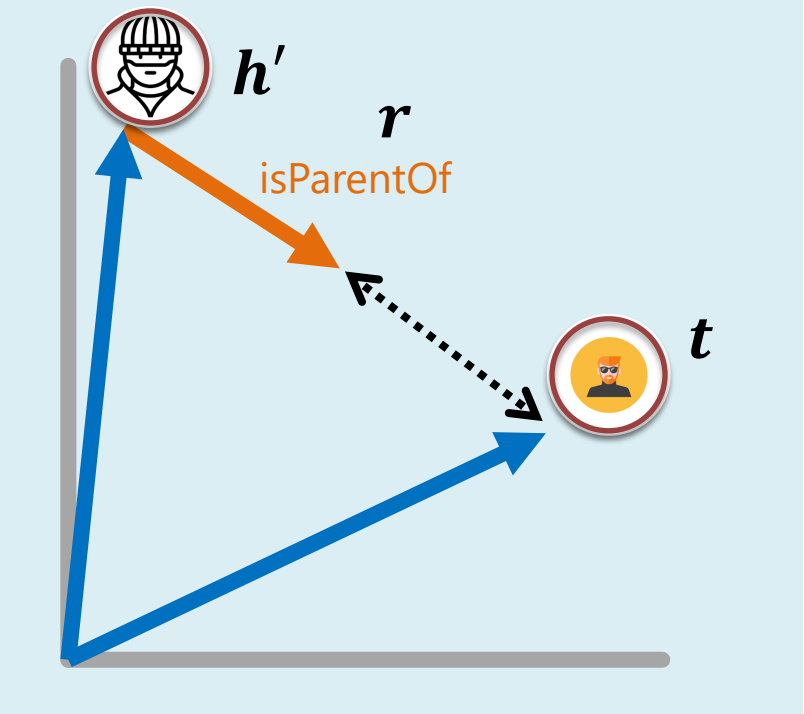
E'

$$\mathcal{L} = \sum_{a \in E} \sum_{a' \in E'_a} [\gamma + f(\boldsymbol{a}) - f(\boldsymbol{a}')]_+$$

a



a'



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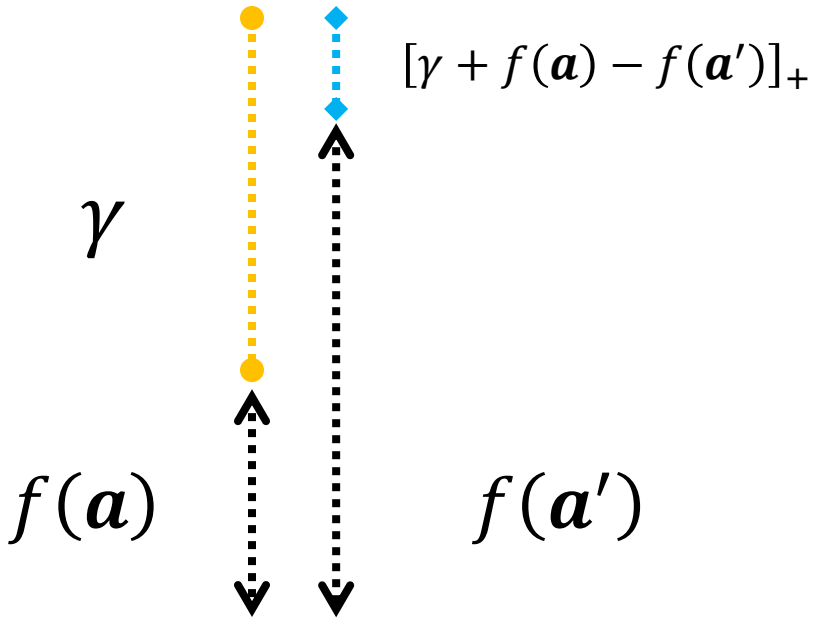

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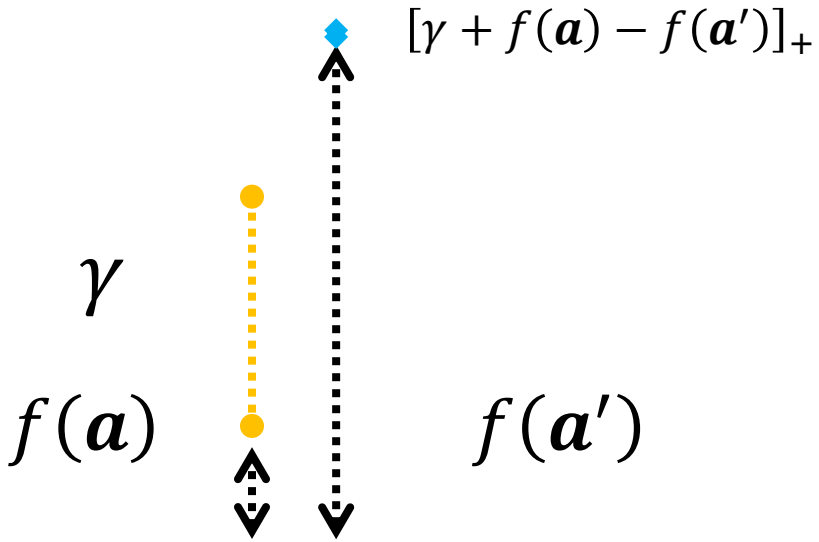

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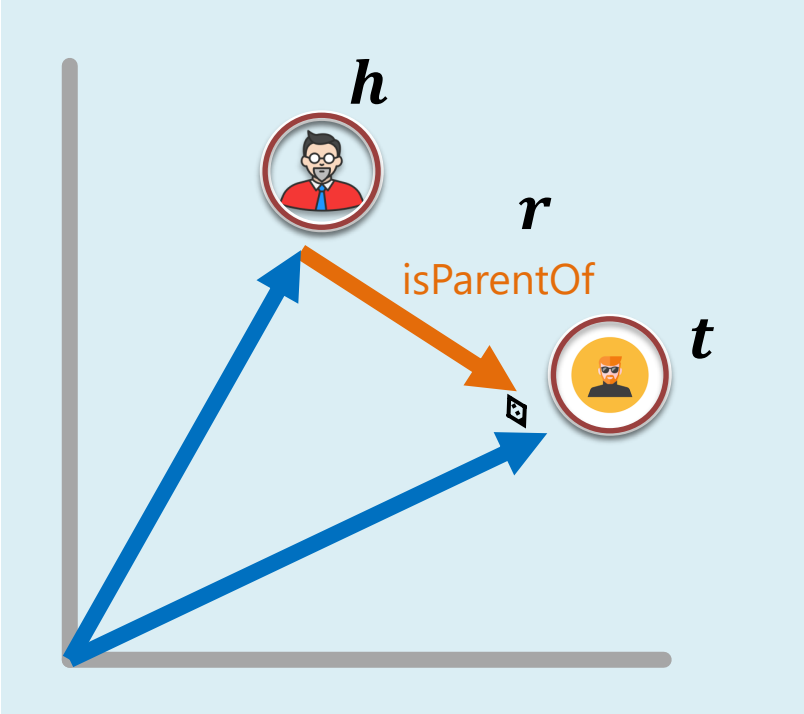
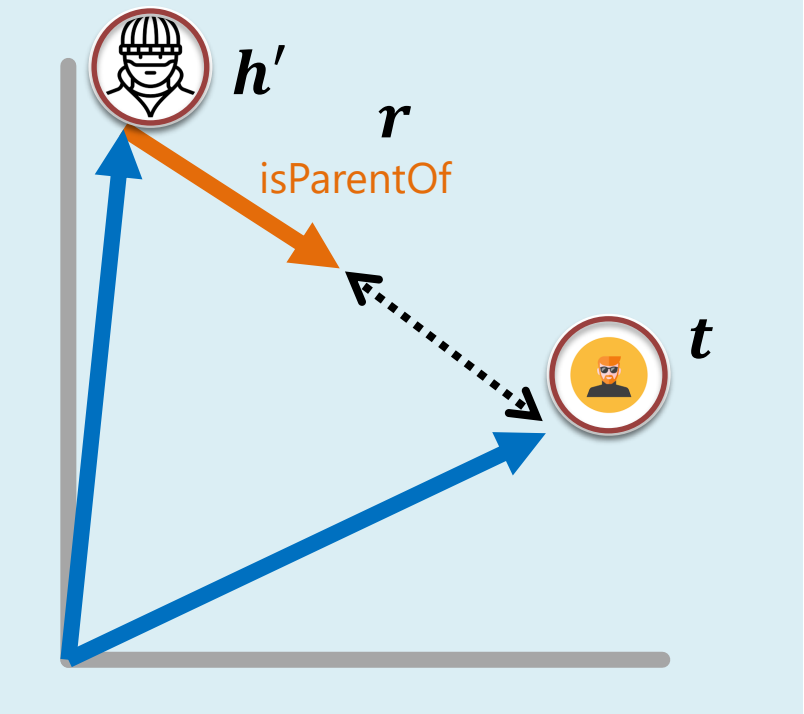
a



a'



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Algorithm uses specific notation!

Algorithm 1 Learning TransE

input Training set $S = \{(h, \ell, t)\}$, entities and rel. sets E and L, margin γ, embeddings dim. k.

initialize $\boldsymbol{\ell} \leftarrow \text{uniform}(-\frac{6}{\sqrt{k}}, \frac{6}{\sqrt{k}})$ for each $\ell \in L$

$\boldsymbol{\ell} \leftarrow \boldsymbol{\ell} / \|\boldsymbol{\ell}\|$ for each $\ell \in L$

$\mathbf{e} \leftarrow \text{uniform}(-\frac{6}{\sqrt{k}}, \frac{6}{\sqrt{k}})$ for each entity $e \in E$

loop

  $\mathbf{e} \leftarrow \mathbf{e} / \|\mathbf{e}\|$ for each entity $e \in E$

  $S_{batch} \leftarrow$ sample(S, b) // sample a minibatch of size b

  $T_{batch} \leftarrow \emptyset$ // initialize the set of pairs of triplets

  for $(h, \ell, t) \in S_{batch}$ **do**

    $(h', \ell, t') \leftarrow$ sample$(S'_{(h,\ell,t)})$ // sample a corrupted triplet

    $T_{batch} \leftarrow T_{batch} \cup \{((h, \ell, t), (h', \ell, t'))\}$

  end for

  Update embeddings w.r.t. $\sum_{\left((h,\ell,t),(h',\ell,t')\right) \in T_{batch}} \nabla \left[\gamma + d(\boldsymbol{h} + \boldsymbol{\ell}, \boldsymbol{t}) - d(\boldsymbol{h'} + \boldsymbol{\ell}, \boldsymbol{t'})\right]_+$

end loop

Source: Antoine Bordes, Nicolas Usunier, Alberto García-Durán, Jason Weston, Oksana Yakhnenko: Translating Embeddings for Modeling Multi-relational Data. NIPS 2013: 2787-2795

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Random (smart) initialization

loop

$\mathbf{e} \leftarrow \mathbf{e} / \|\mathbf{e}\|$ for each entity $e \in E$

Vertex normalization

$S_{batch} \leftarrow \text{sample}(S, b)$ // sample a minibatch of size b

Positive sample

$T_{batch} \leftarrow \emptyset$ // initialize the set of pairs of triplets

for $(h, \ell, t) \in S_{batch}$ **do**

$(h', \ell, t') \leftarrow \text{sample}(S'_{(h,\ell,t)})$ // sample a corrupted triplet

$T_{batch} \leftarrow T_{batch} \cup \{((h, \ell, t), (h', \ell, t'))\}$

Negative sample

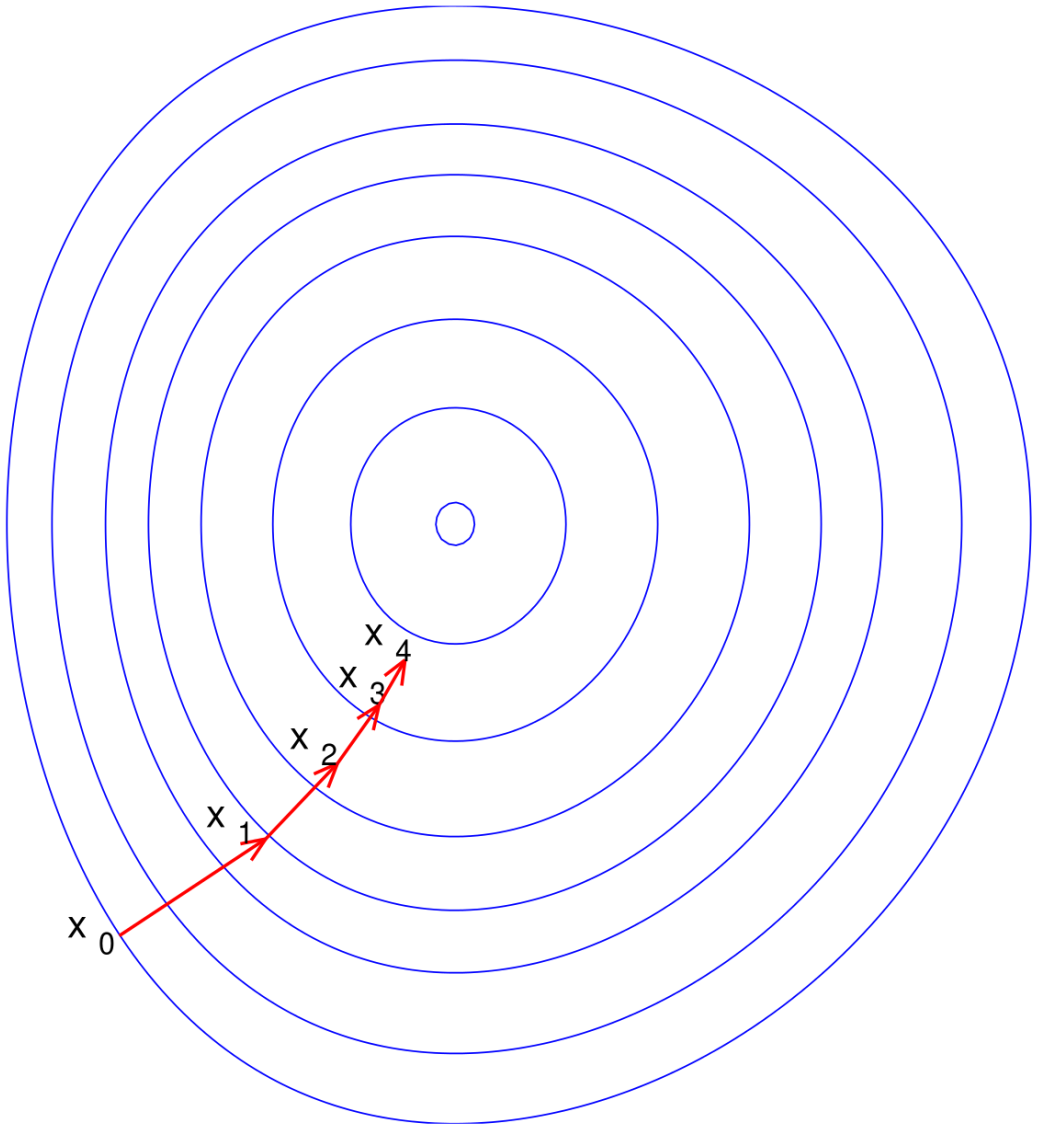
end for

Update embeddings w.r.t. $\sum_{((h,\ell,t),(h',\ell,t')) \in T_{batch}} \nabla \left[\gamma + d(\boldsymbol{h} + \boldsymbol{\ell}, \boldsymbol{t}) - d(\boldsymbol{h'} + \boldsymbol{\ell}, \boldsymbol{t'})\right]_+$

Stochastic Gradient Descent update phase

end loop

Source: Antoine Bordes, Nicolas Usunier, Alberto García-Durán, Jason Weston, Oksana Yakhnenko: Translating Embeddings for Modeling Multi-relational Data. NIPS 2013: 2787-2795



Source: https://en.wikipedia.org/wiki/Gradient_descent#/media/File:Gradient_descent.svg

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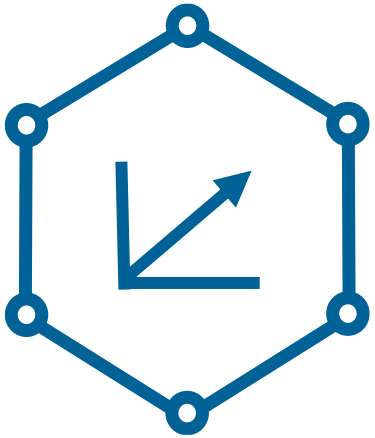

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TransE: Learning

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